

Multivariable Calculus

Unit 4
Jaysen Tsao

4	Multivariate Integrals	2
4.1	Double Integrals	2
	Limit Definition of a Double Integral	2
4.2	Partial Integration	3
4.3	Interpretations of Double Integrals	3
	Double Integrals over General Regions	3
	Double Integrals in Polar Coordinates	3
4.4	Applications of Double Integrals	3
	Surface Area	3
4.5	Triple Integrals	3
4.6	Interpretations of Triple Integrals	3
	Triple Integrals in Cylindrical Coordinates	3
	Triple Integrals in Spherical Coordinates	3
4.7	The Jacobian	3

4 Multivariate Integrals

4.1 Double Integrals

- **Recall:** If we have a function $f(x)$, we can find the area “under the curve” over the interval $[a, b]$ by computing the integral $\int_a^b f(x) dx$.
- Now, imagine that we have a function with two independent variables $f(x, y)$. In a three-dimensional coordinate plane, the function f would describe a surface.
- A **double integral** can be used to find the *volume* “under the surface” defined by f .

Limit Definition of a Double Integral

Recall: The limit definition of the definite integral of $f(x)$ from $x = a$ to $x = b$ is:

$$\int_a^b f(x) dx = \lim_{n \rightarrow \infty} \sum_{k=0}^n f(x_k) \Delta x$$

where $\Delta x = \frac{b-a}{n}$ and $x_k = a + k\Delta x$

Limit Definition of a Double Integral

$$\iint_R f(x, y) dA = \lim_{m, n \rightarrow \infty} \sum_{i=0}^m \sum_{j=0}^n f(x_{ij}^*, y_{ij}^*) \Delta A_{ij}$$

4.2 Partial Integration

4.3 Interpretations of Double Integrals

Double Integrals over General Regions

Double Integrals in Polar Coordinates

4.4 Applications of Double Integrals

Surface Area

4.5 Triple Integrals

4.6 Interpretations of Triple Integrals

Triple Integrals in Cylindrical Coordinates

Triple Integrals in Spherical Coordinates

4.7 The Jacobian